

Nernst Approximation formula

In the context of classical thermodynamics, the Nernst approximation formula is related to the Nernst Heat theorem which is the part of third law of thermodynamics.

The theorem deals with the behaviour of entropy as temperature approaches absolute zero.

One form of Nernst approximation formula is as;

$$\Delta S \approx \frac{\Delta H}{T}$$

ΔS is change in entropy

ΔH is change in enthalpy

T is the absolute Temperature.

This approximation becomes particularly useful at low temp.

As temp. approaches zero the entropy (S) for processes involving pure crystalline substances also becomes zero. This is consistent with the Nernst heat theorem which states that (enthalpy) entropy of any isothermal process approaches zero as temp. approaches zero.

This approximation formula helps describe how entropy changes in a system, especially at low temperatures, aligning with the rules of thermodynamics.

Derivation;

To derive the Nernst approximation formula, the following equations are used,

Entropy change in a process;

For a reversible process, the change in entropy (ΔS) is given as

$$\Delta S = \int \frac{dQ_{\text{rev}}}{T}$$

Here, dQ_{rev} is the infinitesimal heat absorbed reversibly.

Relation between enthalpy and entropy;

For a process occurring at constant pressure, the change in enthalpy is related to heat exchanged:

$$\Delta H = Q_p$$

Therefore for reversible process at constant pressure;

$$\Delta S = \int \frac{d(\Delta H)}{T}$$

low temperature behaviour;

At low temperatures, the entropy change for many processes becomes very small.

The Nernst Heat theorem implies that as $T \rightarrow 0$, ΔS also approaches zero. considering a small temp. change, from T to dT ,

$$\Delta S \approx \frac{\Delta H}{T}$$

This approximation is based on the assumption that the change in enthalpy does not vary significantly over a small temperature range considered.

Application of Nernst Approximation Formula,

This formula is particularly useful at lower temperatures. It allows us to approximate the entropy change for processes involving solids or liquids, where change in heat capacity or phase change may occur.

Here's an example;

Entropy change of a solid at low temperatures;

For a solid at low temperatures where heat capacity is nearly constant, the enthalpy change can be expressed as

$$\Delta H = C_p \Delta T$$

Therefore entropy change can be approximated as

$$\Delta S \approx \frac{C_p \Delta T}{T}$$

If we consider temp. change from T_1 to T_2 then;

$$\Delta S \approx C_p \ln\left(\frac{T_2}{T_1}\right)$$

Conclusion;

The Nernst approximation formula provides a way to estimate the entropy change in a system at low temp, aligning with third law of thermodynamics; emphasizing the behaviour of system at extremely low temp and tendency towards minimal entropy change.